where the weights in each asset class, w_i, apply at the beginning of each period and the returns generated by each asset class, r_i, are realized at the end of the period. The sums are taken across asset classes indexed by i. The superscript bs denote the weights (w_i^b) and returns (r_i^b) of the asset classes in the benchmark.

Equation (14.3) has partitioned the active returns into a timing and selection component. One advantage of this decomposition is that we can assign each separate source of return to different responsible parties. In Norway's case, the Ministry of Finance is responsible for the passive benchmark decision. NBIM, the fund manager, is responsible for the active decision away from the benchmark. According to equation (14.3), NBIM can generate positive active returns by timing the allocations between equities and bonds or by choosing securities that outperform within the equities or fixed income asset classes.

Here's a return decomposition for a fund concerned about returns in excess of inflation:

$$\underbrace{r-\pi}_{\text{Real Return}} = \underbrace{(r-r_f)}_{\text{Risk Premium}} + \underbrace{(r_f-\pi)}_{\text{Real Cash Return}}, \tag{14.4}$$

where π is the inflation rate and r_f is the risk-free return on U.S. T-bills. A real return can also be decomposed using a benchmark, r_{bmk}, with asset class weights w_i and returns r_i, similar to equation (14.3):

$$\begin{aligned} r-\pi = & \underbrace{(r_f-\pi)}_{\text{Real Cash Return}} + \underbrace{\sum w_i^b(r_i^b - r_f)}_{\text{Strategic Asset Allocation}} \\ & + \underbrace{\sum (w_i - w_i^b) r_i^b}_{\text{Tactical Asset Allocation}} + \underbrace{\sum w_i(r_i - r_i^b)}_{\text{Security Selection}}. \end{aligned} \tag{14.5}$$

If we aim to beat inflation, we start with the real cash return in equation (14.5), which is the difference between the nominal risk-free return and inflation. Then strategic asset allocation aims to pick asset classes that produce returns in excess of the nominal risk-free return, and it is now an active decision.[9] In Norway's case, the strategic asset allocation is the constant equity-bond mix. For other funds, the strategic asset allocation slowly varies over time. The tactical asset allocation and security selection components are the same as the last two terms in equation (14.3). Even though both the strategic asset allocation and tactical asset allocation involve changing asset class weights, practitioners use the term "tactical" to distinguish the shorter-term changes in the asset class holdings from the more slowly moving strategic allocations.

[9] Chapter 4 interprets strategic and tactical allocation in the context of long-run portfolio choice. Equation (14.5) is stated as a return in excess of inflation. When we replace inflation with a liability, the decomposition reflects a *surplus* return, or a return in excess of liabilities.